

Invariant Learning on Heterogeneous Graphs via Subgraph Environment Inference

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Abstract

The out-of-distribution (OOD) generalization of graph neural networks poses significant challenges in Web applications, where data resides in complex heterogeneous information networks. Such networks exhibit not only structural heterogeneity but also distribution shifts arising from evolving user behaviors and data collection biases. Conventional GNNs often struggle to identify stable patterns in such heterogeneous graphs, particularly in the absence of explicit environment labels. The core issue is that latent environments can create spurious correlations between node features, local topology, and labels. Models may then rely on these environment-specific shortcuts for predictions, failing to learn the invariant mechanisms that generalize under distribution shifts. To address these limitations, we propose InvHG (Invariant Learning on Heterogeneous Graphs via Subgraph Environment Inference), a causality-inspired framework that infers latent environments at the subgraph level, disentangles type-specific confounding effects, and leverages regularized expert fusion to learn invariant representations. Extensive

experiments on heterogeneous graph OOD benchmarks demonstrate that InvHG consistently outperforms state-of-the-art methods, offering a robust solution for complex Web graph learning. The source code is available at <https://github.com/mok630/InvHG>.

CCS Concepts

• Computing methodologies → Learning paradigms.

Keywords

Heterogeneous Graph Neural Networks, Out-of-Distribution Generalization, Environment Inference, Invariant Learning

ACM Reference Format:

Yanghui Fu, Yunfei Wang, Hao Zou, Yue He, Haotian Wang, Qing Cheng, Guangquan Cheng, and Shixuan Liu. 2026. Invariant Learning on Heterogeneous Graphs via Subgraph Environment Inference. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792342>

1 Introduction

The modern Web relies on massive-scale relational data from social media, e-commerce, and knowledge bases, which is inherently heterogeneous with diverse entities and relations. To capture this rich semantic structure, heterogeneous graphs (HGs) [48] have become a key paradigm, used in Web systems like social networks [7, 28, 49] and recommendation systems [4, 13, 43], as

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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792342>

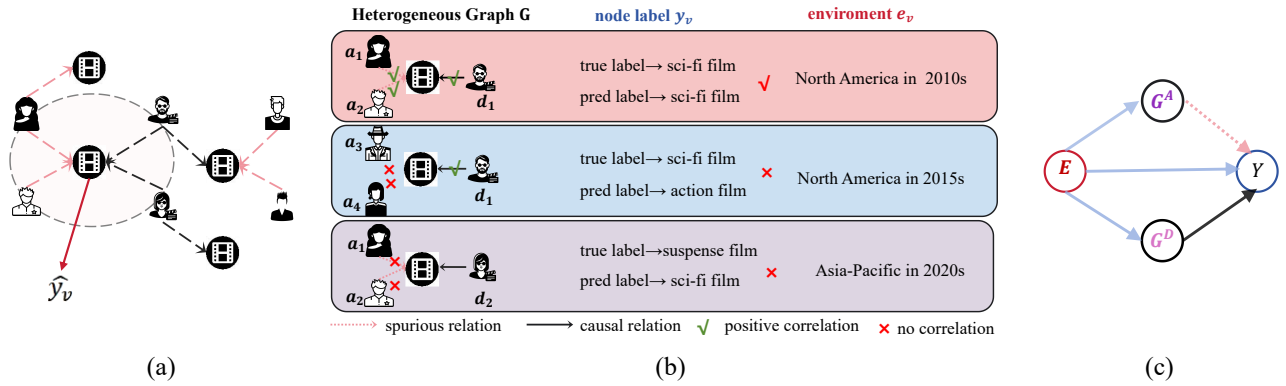


Figure 1: An illustration based on a heterogeneous movie graph for node-label prediction with HGNNs. (a) The task aims at predicting the target node's label $\hat{y}_v$. (b) Environments modulate generalization (examples across regions/years). Director–movie edges encode stable causal signals, whereas actor–movie edges capture environment-sensitive shortcuts. (c) A structural causal model showing dependencies among the actor subgraph G^A , the director subgraph G^D , the label Y , and the unobserved environment E ; E acts as a latent confounder—a common cause of G and Y in data generation. Assumption: directors' styles and genre preferences are relatively stable over time, thus $G^D \rightarrow Y$ is expected to remain invariant across environments.

well as other domains such as biological pathways [1, 14, 21]. Unlike their homogeneous counterparts, HGs explicitly preserve the diverse node and edge types, providing more powerful representations for downstream tasks like node classification [12, 45]. To model these structures, Heterogeneous Graph Neural Networks (HGNNs) [19, 35, 39]—a specialized class of GNNs [16, 23, 38]—have emerged as state-of-the-art encoders that propagate information over typed structures to effectively learn node representations.

Despite these advances, existing HGNNs and their homogeneous graph counterparts typically rely on a key assumption: that the training and testing data are drawn from the same distribution. In real-world scenarios, however, distribution shifts are inevitable due to evolving data generation mechanisms [24, 40], such as changes in user demographics, temporal effects in systems, or regional contextual differences. These shifts pose severe challenges for OOD generalization: HGNNs often overfit to spurious correlations, leading to significant performance degradation. Existing evidence indicates that GNNs are prone to capturing spurious environmental correlations in the data rather than stable causal relationships [41, 42].

Consider Figure 1(a): a movie HG with movies (M), actors (A), and directors (D) where the task is genre prediction from structure. Director–movie paths encode *stable* signals: a director's style/genre preference tends to persist across environments. By contrast, actor–movie paths can be *environment-sensitive* shortcuts, because actors' participation patterns often track transient market and production factors (e.g., genre cycles, star branding, casting practices, and regional tastes) that vary across eras and regions, e.g., in North America in the 2010s, many actors clustered in sci-fi, so strong ties to those actors spuriously predict sci-fi. When environments shift—North America in the mid-2010s as actors diversify, or Asia-Pacific in the 2020s with suspense trending—such shortcuts fail (Figure 1(b)). The structural causal model (SCM) in Figure 1(c) formalizes this: latent environment E (era/region trends) affects both graph structure G (actor–movie G^A , director–movie G^D) and labels Y , so HGNNs may amplify environment-sensitive paths, whereas director–movie signals remain more robust across environments.

As shown, OOD in HGs is intrinsically complex. Unlike homogeneous graphs, these shifts are type-conditioned, originating from specific node and edge types and propagating along semantic meta-paths, which makes true global invariances hard to identify. Existing graph OOD research has largely centered on homogeneous graphs. In particular, a major strand is invariant learning [20, 32], which seeks predictors whose behavior remains stable across multiple environments, usually by grouping data with environment labels and penalizing environment-specific variations (e.g., IRM) or optimizing worst-group risk (e.g., Group-DRO). However, real-world graphs seldom provide such labels, especially in heterogeneous settings where environment factors are intertwined with type and path-level semantics. As a result, invariant-learning techniques are difficult to apply directly, which complicates causal identification and intervention: confounders remain entangled and models are prone to learn spurious rather than causal relations.

Recent work on environment inference in homogeneous graphs attempts to address this gap by inferring latent environments and applying interventions to mitigate unobserved confounding [41]. However, directly applying such methods to HGs faces two key challenges due to heterogeneity: (1) Fine-grained confounding identification: HGs exhibit multi type-specific confounding where different node types are governed by distinct environmental factors. Homogeneous graph-based methods typically use a single global environment estimator, failing to capture type-specific confounding and thus misidentifying confounders. (2) Semantics-aware message passing: Structural shifts in HGs are semantics-dependent, concentrating within specific edge-type subgraphs. Approaches designed for homogeneous graphs ignore edge-type semantics during message passing, conflating structural shifts across semantic spaces and amplifying distributional drift through misguided aggregation. These challenges call for a heterogeneity-aware causal model that captures type-conditioned environments and path-level stability.

To address these issues, we propose InvHG (Invariant Learning on Heterogeneous Graphs via Subgraph Environment Inference), an environment inference for invariant learning framework for OOD generalization in HGs. Drawing from causal analysis, we

model data generation via a SCM where latent environments E confound the graph features G and labels Y (Figure 1(c)). Invariant principles posit that predictions focusing on environment-agnostic paths remain robust [2, 29]. Specifically, InvHG comprises three key components: a type-aware environment encoder, which equips each node type with an independent inference module to project nodes into type-specific pseudo-environment spaces by leveraging causal-inspired subgraphs, finely characterizing differentiated confounding influences; a subgraph mixture-of-experts layer, which designs expert networks for each edge type and utilizes pseudo-environments for dynamic routing of messages, adaptively selecting stable propagation modes; and an adaptive subgraph fusion aggregator, which employs node-level weighting mechanisms to fuse multi-subgraph outputs, achieving cross-type global invariant representations. Through environment routing and robust fusion, InvHG severs confounding paths, guiding the model to learn causal relationships from heterogeneous structures to labels. We conduct extensive experiments on multiple real-world heterogeneous graph datasets, encompassing various shift. Results demonstrate that InvHG consistently outperforms existing homogeneous and heterogeneous graph OOD baselines, achieving new state-of-the-art performance on OOD test sets.

In summary, the key contributions of our work are: (1) a systematic analysis of the unique challenges in HG OOD generalization, revealing the role of typed confounding factors and extending causal interventions to heterogeneous settings; (2) the proposal of InvHG, an end-to-end invariant learning framework that addresses type-specific biases; (3) comprehensive experiments validating its effectiveness and robustness across diverse shifts, advancing the frontiers of heterogeneous graph OOD research.

2 Related Work

Heterogeneous Graph Neural Networks. Traditional graph neural networks such as GCN [23] and GAT [38] are designed under the assumption of homogeneous graph structures, where all nodes and edges share the same type. This assumption restricts their applicability in HGs that involve multiple node and relation types. To address this challenge, a series of heterogeneous GNN models have been proposed, including HeteroGCN [48], HeteroGraphSAGE [8], HeteroGAT [39], and HGT [19]. These methods extend convolution and aggregation mechanisms with type-specific designs, thereby capturing both structural and semantic heterogeneity for more effective representation learning.

OOD Generalization on Graphs. Learning under distribution shifts has recently become a central research topic in graph machine learning, since graph-structured data often arises from diverse environments with changing semantics and connectivity patterns. Existing works on graph OOD generalization can be broadly categorized into two major lines of research: **(1) Invariant learning.** This direction seeks to identify predictive patterns that remain stable across environments, often inspired by causality or invariant risk minimization [3, 25, 34]. Representative methods include DIR [44], CIGA [5], CAL [36], StableGNN [11], CaNet [41] and so on [6, 9, 10, 46, 50]. Despite their effectiveness, most invariant-based methods are designed for homogeneous graphs and focus on graph classification, leaving node property prediction on heterogeneous graphs largely unexplored. **(2) Data augmentation and**

consistency. Another line of works improves OOD robustness by generating diverse training environments or enforcing multi-view consistency. In graph learning, this includes edge/node perturbations, subgraph sampling, adversarial or contrastive regularization, and mixup-based strategies [17, 26, 27, 31, 37, 47]. Nevertheless, such augmentations lack causal guarantees and may introduce semantic distortions; moreover, they often ignore relation-type sensitivity, limiting their effectiveness in heterogeneous graphs.

3 Problem Formulation

Heterogeneous graph. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi, \psi)$ be a heterogeneous graph, where $\mathcal{V}$ and $\mathcal{E}$ denote the node and edge sets, respectively. Each node $v \in \mathcal{V}$ is associated with a node type $\varphi(v) \in \mathcal{T}$, and each edge $(u, v) \in \mathcal{E}$ is associated with a relation type $\psi(u, v) \in \mathcal{R}$, where $\mathcal{T}$ and $\mathcal{R}$ denote the sets of node types and relation types. Let $\Phi \in \mathcal{T}$ be the *target* node type for classification. We denote node features by $X \in \mathbb{R}^{|\mathcal{V}| \times D}$ and write A for the typed adjacency representation of $\mathcal{G}$. Table 1 summarizes notations frequently used in this paper for quick reference.

Table 1: Notations used in this paper.

Notation	Explanation
$\mathcal{G}$	Heterogeneous graph
$\mathcal{V}$	Set of nodes
$\mathcal{E}$	Set of edges
$\varphi(v)$	Node-type mapping function, $\varphi : \mathcal{V} \rightarrow \mathcal{T}$
$\psi(u, v)$	Edge-type mapping function, $\psi : \mathcal{E} \rightarrow \mathcal{R}$
$\mathcal{T}$	Set of node types
$\mathcal{R}$	Set of relation (edge) types
Φ	Target node type for classification
$X \in \mathbb{R}^{ \mathcal{V} \times D}$	Node feature matrix (D = feature dimension)
A	Typed adjacency representation of $\mathcal{G}$
$z_v^{[l]}$	Representation of node v at layer l
$e_v^{[l], r}$	Pseudo-environment embedding for node v under relation r at layer l
r_r	Learnable embedding vector of relation type $r \in \mathcal{R}$

Node Classification on HGs. For each target-type node v with $\varphi(v) = \Phi$, let $y_v \in \{1, 2, 3, \dots, C\}$ be its label, where C is the number of classes. Given a set of labeled training target nodes $\mathcal{V}_{tr} \subset \mathcal{V}$ with $\{y_v\}_{v \in \mathcal{V}_{tr}}$ and the full graph (X, A) , an HGNN learns a predictor

$$f_\theta : (\mathcal{G}, v) \mapsto \hat{y}_v \in \Delta^C, \quad (1)$$

parameterized by θ , that infers labels for test nodes $\mathcal{V}_{te} = \mathcal{V} \setminus \mathcal{V}_{tr}$ using the multi-typed relational context in $\mathcal{G}$. The standard training objective is empirical cross-entropy:

$$\theta^* = \arg \min_{\theta} \frac{1}{|\mathcal{V}_{tr}|} \sum_{v \in \mathcal{V}_{tr}} \mathcal{L}(y_v, f_\theta(\mathcal{G}, v)), \quad (2)$$

OOD Generalization on HGs. In real-world scenarios, the data distribution is not fixed but shifts across environments due to factors such as temporal dynamics, user demographics, or regional contexts. Let E denote a random variable representing the environment, and e be its realization. The joint distribution of graphs and labels can be characterized as $P(G, Y|E) = P(G|E)P(Y|G, E)$, where G is the heterogeneous graph structure, and Y is the label of nodes. This formulation indicates that the environment E acts as a latent confounder, simultaneously influencing both the graph structure G (e.g., actor–movie, or director–movie connections in a

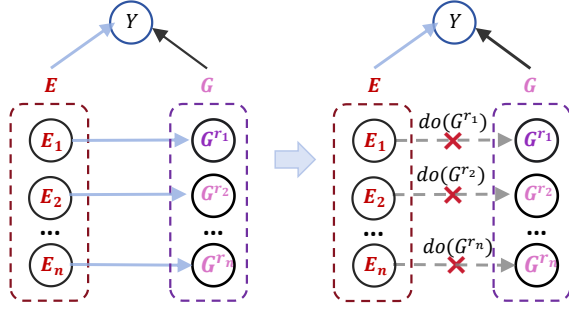


Figure 2: Structural Causal Model. G^{r_i} means different type-specific subgraph, E_i means environment of subgraph G^{r_i}

classification) and the label Y (e.g., movie genre). The goal of OOD generalization is to train a model on a set of environments E_{tr} such that it can generalize to unseen environments E_{te} , even when the training and testing distributions differ, i.e., $P_{tr}(G, Y) \neq P_{te}(G, Y)$. A key challenge is that models trained via maximum likelihood estimation tend to capture spurious, environment-sensitive correlations rather than stable causal relations, which leads to performance degradation when deployed in new environments.

4 Proposed Model

We next present our analysis and proposed method. In Section 4.1, we first analyze the generalization behaviors of common HGNNs in node property prediction under distribution shifts and reveal what causes the deficiency of HGNNs w.r.t. out-of-distribution data, with a focus on the unique challenges in heterogeneous graphs. Based on the analysis, in Sections 4.2 and 4.3, we introduce the formulation and instantiations for our proposed model, respectively.

4.1 Causal Analysis for HGNNs

For a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi, \psi)$, HGNNs iteratively update node representations through type-aware aggregation of neighbored nodes' features, conditioned on target node and edge types. Specifically, assume $z_v^{(\ell)}$ as the representation of node v at the ℓ -th layer. The updating rule of common HGNNs can be written as

$$z_v^{(\ell+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \text{Conv}_r^{(\ell)} \left(\{z_u^{(\ell)} \mid u \in \mathcal{N}_v^{(r)} \cup \{v\}\} \right) \right), \quad (3)$$

where $\mathcal{N}_v^{(r)}$ is the set of r -type neighbors of v , $\text{Conv}_r^{(\ell)}$ is a type-specific graph convolution operator, and σ is an activation function. Thus, the prediction is $\hat{y}_v = f_\theta(\mathcal{G})$, where f_θ denotes the HGNN's parameters θ . The objective minimizes cross-entropy over training nodes $\mathcal{V}_{tr}$:

$$\theta^* = \arg \min_{\theta} - \frac{1}{|\mathcal{V}_{tr}|} \sum_{v \in \mathcal{V}_{tr}} y_v^\top \log f_\theta(\mathcal{G}). \quad (4)$$

To understand the OOD generalization failure of HGNNs, we introduce a causal perspective. We postulate a latent *environment variable* E , which encompasses all exogenous factors causing spurious correlations between the graph $\mathcal{G}$ and labels $\mathcal{Y}$. For instance, in movie recommendation, E could represent regional demographics that influence both casting choices ($\mathcal{G}$) and genre popularity ($\mathcal{Y}$).

This leads to the causal graph depicted in Figure 2: **1) Causal Path:** $\mathcal{G} \rightarrow \hat{\mathcal{Y}}$. This is the desired mechanism where predictions are based on the genuine semantic information in the graph. **2) Confounding Paths:** $E \rightarrow \mathcal{G}$ and $E \rightarrow \mathcal{Y}$. This indicates that the environment E biases both the graph structure and the node labels. During training, the objective $p_\theta(\hat{\mathcal{Y}}|\mathcal{G})$ inadvertently captures the spurious association along the back-door path $\mathcal{G} \leftarrow E \rightarrow \hat{\mathcal{Y}}$, making the model reliant on environment-sensitive features.

In heterogeneous graphs, this confounding is exacerbated and becomes *type-specific*. We can partition E into components $E_{i=1}^n$, each governing a specific relation type, leading to subgraph-specific dependencies $\mathcal{G}^i \leftarrow E_i \rightarrow \mathcal{Y}$. This type-specific confounding operates at two interconnected levels: (1) *Multi-type Confounding*: At the environmental level, different semantic types (e.g., actors, directors) are driven by distinct factors E_i (e.g., casting trends, directorial styles), creating multiple, independent sources of bias. (2) *Semantics-dependent Shifts Under Subgraphs*: At the graph level, these environmental influences manifest as systematic variations in edge distributions within each relation type r , directly perturbing the type-aware message passing that is fundamental to HGNNs. Consequently, an HGNN trained via $p_\theta(\hat{\mathcal{Y}}|\mathcal{G})$ learns to exploit these *type-specific shortcuts*—statistical regularities present in the training environment $p_{tr}(E)$ but unstable under $p_{te}(E) \neq p_{tr}(E)$, leading to OOD performance degradation.

The causal view prescribes a direct solution: to deconfound the training by modeling the interventional distribution $p_\theta(\hat{\mathcal{Y}} \mid do(\mathcal{G}))$. The *do*-operator severs the incoming edges to $\mathcal{G}$ in the causal graph (i.e., $E \rightarrow \mathcal{G}$), thereby blocking the confounding back-door path. This forces the model to learn based only on the causal relationship $\mathcal{G} \rightarrow \hat{\mathcal{Y}}$, which is stable across environments. Our goal is thus to develop a method that approximates this ideal interventional learning within the HGNN framework.

4.2 A Tractable Causal Estimator for HGs

An ideal computation of $p_\theta(\hat{\mathcal{Y}} \mid do(\mathcal{G}))$ requires physical intervention on $\mathcal{G}$ (e.g., randomized trials over typed subgraphs), which is intractable. We thus resort to observational data-based learning via backdoor adjustment [25], yielding (see Appendix A.1)

$$p_\theta(\hat{\mathcal{Y}} \mid do(\mathcal{G})) = \mathbb{E}_{p_0(E)} [p_\theta(\hat{\mathcal{Y}}|\mathcal{G}, E)], \quad (5)$$

where $p_0(E)$ is a prior over environments, reflecting plausibility without data bias. Instantiating $p_\theta(\hat{\mathcal{Y}}|\mathcal{G}, E)$ is challenging due to unknown E . Even partial proxies (e.g., timestamps in citation networks) are insufficient for typed confounding.

We introduce *relation-wise* pseudo-environments to disentangle stable and environment-sensitive signals in heterogeneous graphs. Concretely, for each relation $r \in \mathcal{R}$ and destination node v , a type-aware estimator produces a pseudo-environment $\mathbf{e}_v^{[l],r}$ from the typed subgraph $\mathcal{G}^{(r)}$:

$$q_\phi(E^r \mid \mathcal{G}^{(r)}, v, z_v^{[l]}) \Rightarrow \mathbf{e}_v^{[l],r},$$

where the estimator is type-specific w.r.t. $\varphi(v)$ to capture differentiated confounding across node types. Conditioned on the per-relation type dictionary $E^r = \{\mathbf{e}_\tau^r\}_{\tau \in \mathcal{T}_r}$, the HGNN predictor models $p_\theta(\hat{\mathcal{Y}} \mid \{\mathcal{G}^{(r)}, E^r\}_{r \in \mathcal{R}})$.

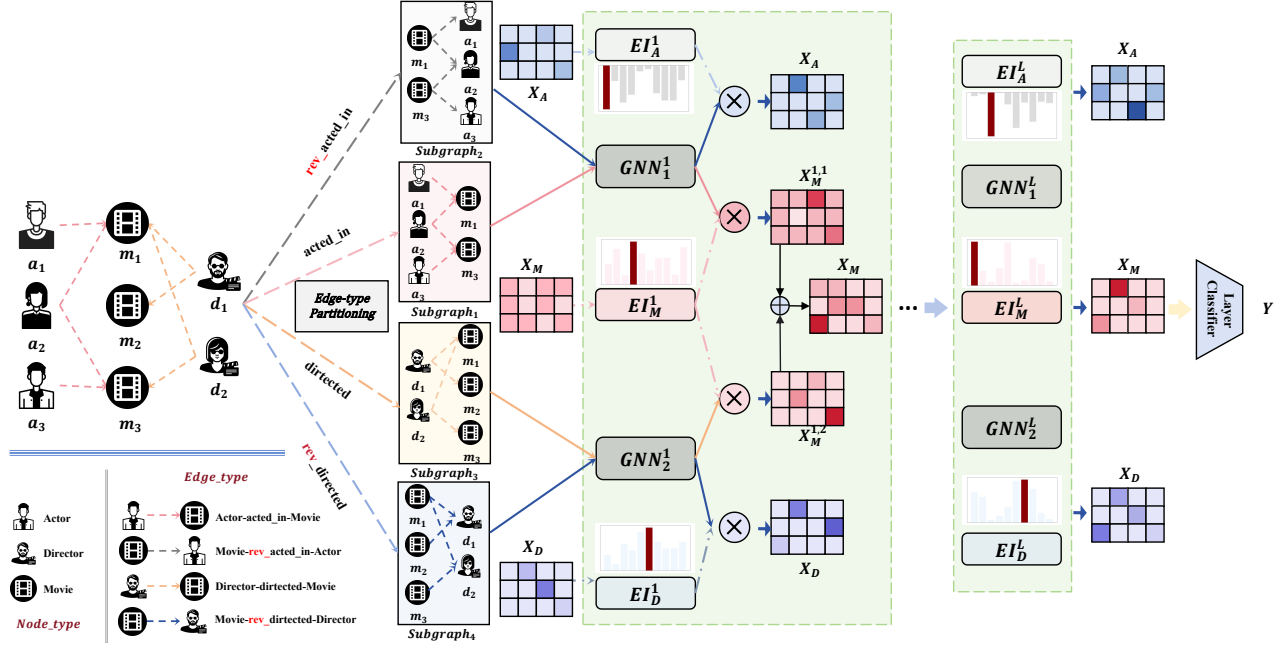


Figure 3: InvHG: Overall framework. The model achieves OOD-robust node prediction on a heterogeneous graph by (i) *edge-type partitioning*, (ii) *type-aware environment inference*, and (iii) *relation-wise encoding and fusion*, repeated over L layers before classification. Node types $\mathcal{T} = \{\text{Actor (A), Movie (M), Director (D)}\}$; relation (edge) types $\mathcal{R}$ (e.g., actor $\rightarrow$ acted_in $\rightarrow$ movie, director $\rightarrow$ directed $\rightarrow$ movie, and their reverses); typed subgraph for relation r : $\mathcal{G}^{(r)}$ (*Subgraph_r*); node-feature matrices X_A, X_M, X_D ; type-specific encoder at layer l : $\text{GNN}_l^{[l]}$; environment inference module for node type $\tau \in \mathcal{T}$ at layer l : $\text{EI}_\tau^{[l]}$; relation-wise pseudo-environment for node v : $\mathbf{e}_v^{[l],r}$; fusion operators $\oplus$ (aggregation across relations) and $\times$ (environment-conditioned weighting); target node type for prediction Φ (e.g., movies); final label Y .

Under a factorized variational family $q_\phi(\{E^r\}_{r \in \mathcal{R}} | \mathcal{G}) = \prod_{r \in \mathcal{R}} q_\phi(E^r | \mathcal{G}^{(r)})$ (message-passing receivers) of relation r . This is distinct from the $\mathcal{G}^{(r)}$ and an independent prior $p_0(\{E^r\}) = \prod_{r \in \mathcal{R}} p_0(E^r)$, the interventional objective admits the following variational lower bound (see Appendix A.2):

$$\log p_\theta(\hat{\mathcal{Y}} | \text{do}(\mathcal{G})) \geq \mathcal{L}_{\text{sup}} - \mathcal{L}_{\text{reg}}, \quad (6)$$

$$\mathcal{L}_{\text{sup}} := \mathbb{E}_{\prod_{r \in \mathcal{R}} q_\phi(E^r | \mathcal{G}^{(r)})} [\log p_\theta(\hat{\mathcal{Y}} | \{\mathcal{G}^{(r)}, E^r\}_{r \in \mathcal{R}})], \quad (7)$$

$$\mathcal{L}_{\text{reg}} := \sum_{r \in \mathcal{R}} \text{KL}(q_\phi(E^r | \mathcal{G}^{(r)}) \| p_0(E^r)). \quad (8)$$

4.3 Model Instantiations

We now instantiate the proposed causal treatment for HGs, following the overall architecture illustrated in Figure 3. The computation in each layer involves three key steps: edge-type subgraph partitioning, environment inference, and subgraph fusion.

4.3.1 Edge-type Subgraph Partitioning. Given $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \varphi, \psi)$, we partition it by relation types $r \in \mathcal{R}$:

$$\mathcal{G}^{(r)} = (\mathcal{V}, \mathcal{E}^{(r)}, \varphi, \psi), \quad \mathcal{E}^{(r)} = \{(u, v) \in \mathcal{E} \mid \psi(u, v) = r\}.$$

For every forward edge (u, v) of type r , we also add the reverse edge (v, u) as type r^{-1} to ensure bidirectional information flow, yielding a family $\{\mathcal{G}^{(r)}\}_{r \in \mathcal{R}}$ of typed subgraphs.

4.3.2 Environment Inference in Subgraphs. For each typed subgraph $\mathcal{G}^{(r)}$, we infer a *pseudo-environment* on the destination nodes

classification targets, which are restricted to nodes with $\varphi(v) = \Phi$. Given $z_v^{[l]}$ for a destination node v of type $\tau = \varphi(v)$, an estimator produces logits over K pseudo-environments:

$$\boldsymbol{\pi}_v^{[l],r} = \text{Softmax}(W_{\tau,S}^{[l],r} z_v^{[l]}), \quad W_{\tau,S}^{[l],r} \in \mathbb{R}^{H \times K}. \quad (9)$$

We then obtain an environment embedding $\mathbf{e}_v^{[l],r} \in \mathbb{R}^K$ using the Gumbel–Softmax reparameterization:

$$\mathbf{e}_{vk}^{[l],r} = \frac{\exp((\pi_{vk}^{[l],r} + g_k)/\tau_g)}{\sum_{k'=1}^K \exp((\pi_{vk'}^{[l],r} + g_{k'})/\tau)}, \quad g_k \sim \text{Gumbel}(0, 1), \quad (10)$$

with temperature τ_g . The vector $\mathbf{e}_v^{[l],r}$ softly routes $z_v^{[l]}$ to lightweight experts inside the estimator, capturing environment-dependent variations of relation r while keeping the focus on heterogeneous modeling.

4.3.3 Subgraph Fusion for Node-type Updates. As shown in Eq. (3), for each relation $r \in \mathcal{R}$ the convolution produces a relation-specific representation $z_v^{[l],r}$ for node v . For a node v of type $\tau = \varphi(v)$, we fuse its relation-specific representations $\{z_v^{[l],r}\}_{r \in \mathcal{R}}$ into a global update:

$$\tilde{z}_v^{[l+1]} = \sum_{r \in \mathcal{R}} \alpha_{v,r}^{[l]} z_v^{[l],r}, \quad (11)$$

where the attention weights are

$$\alpha_{v,r}^{[l]} = \frac{\exp(\text{MLP}_\tau(z_v^{[l]})^\top \mathbf{r}_r)}{\sum_{r' \in \mathcal{R}} \exp(\text{MLP}_\tau(z_v^{[l]})^\top \mathbf{r}_{r'})}, \quad \mathbf{r}_r \in \mathbb{R}^H. \quad (12)$$

This fusion integrates multi-relational evidence per node type and reduces over-reliance on unstable, environment-sensitive relations.

4.3.4 Layer Stacking and Prediction. The above steps are repeated for $l = 1, \dots, L$ to refine node embeddings. Finally, for a classification target node v with $\varphi(v) = \Phi$, we predict

$$\hat{y}_v = \phi_{\text{out}}(z_v^{[L]}). \quad (13)$$

This instantiation explicitly models type-specific relations while accounting for latent environments, thereby improving OOD generalization on heterogeneous graphs.

Optimization Objective. We train the model with a node-level likelihood conditioned on all relation-wise subgraphs and the inferred pseudo-environments, together with a light regularization on the environment estimator. For a classification target v with $\varphi(v) = \Phi$, the predictive distribution is $p_\theta(y_v | \{\mathcal{G}^{(r)}, \mathbf{e}_v^{[L],r}\}_{r \in \mathcal{R}})$, where $\mathbf{e}_v^{[L],r}$ is the relation-wise pseudo-environment embedding at the last layer. The overall loss is:

$$\begin{aligned} \mathcal{L}(\theta, \phi) = & \frac{1}{|\mathcal{V}_{\text{tr}}^\Phi|} \sum_{v \in \mathcal{V}_{\text{tr}}^\Phi} \left[-\log p_\theta(y_v | \{\mathcal{G}^{(r)}, \mathbf{e}_v^{[L],r}\}_{r \in \mathcal{R}}) \right] \\ & + \lambda \frac{1}{L|\mathcal{R}|} \sum_{l=1}^L \sum_{r \in \mathcal{R}} \text{KL}(q_\phi(\mathbf{e}_v^{[l],r} | \mathcal{G}^{(r)}) \| p_0(E)). \end{aligned} \quad (14)$$

where $p_0(E)$ is a simple prior over environments, and λ balances the two terms.

Closed-form surrogate for the KL. Let $q_\phi(\cdot)$ produce logits $\pi_v^{[l],r}$ and $\mathbf{e}_v^{[l],r}$ be the (soft) Gumbel-Softmax sample. With a uniform prior $p_0(E) = \text{Cat}(\frac{1}{K}, \dots, \frac{1}{K})$, we use the standard surrogate

$$\text{KL}(q_\phi(\mathbf{e}_v^{[l],r} | \mathcal{G}^{(r)}, v) \| p_0(E^r)) \approx \sum_{k=1}^K e_{vk}^{[l],r} (\log \pi_{vk}^{[l],r} + \log K), \quad (15)$$

Optimization and Complexity. We optimize θ and ϕ jointly using gradient-based methods on Eq. (14). The computational complexity is $O(L\mathcal{K}|\mathcal{S}||\mathcal{E}|)$, where $|\mathcal{S}|$ is the number of subgraphs.

5 EXPERIMENTS

We design a series of experiments on real-world datasets to answer the following research questions: **RQ1 Effectiveness:** How does the InvHG perform compared with other state-of-the-art node classification methods in OOD environments? **RQ2 Ablation:** How much does each component of InvHG contribute to the overall performance? **RQ3 Sensitivity:** How does InvHG perform under different parameter settings?

5.1 Experimental Settings

Dataset. We evaluate on three real-world heterogeneous graphs from distinct domains: TMDB (movies), ArXiv (academic citations), and Book (publishing) [30]. The datasets span a wide range of scales and cover multi-class and multi-label prediction on different target types. Detailed statistics are summarized in Table 5.

OOD Settings. We consider the following two distinct scenarios: **1. Transductive (within-graph).** For each dataset, we follow [15]

to construct OOD splits inside a single graph via two criteria: (i) *year-based* splitting, where earlier years form the training pool and later years are held out as OOD test; (ii) *degree-based* splitting, where high-degree nodes constitute the training pool and low-degree nodes are held out as OOD test. From the training pool we further draw an i.i.d. split of 8:1:1 for train/val/test to report in-distribution performance; **2. Inductive (cross-graph).** We assess cross-environment generalization by temporal slicing into disjoint subgraphs. 1) We create two non-overlapping temporal subgraphs (Graph A / Graph B) and train on A, test on B to measure single-step temporal shift. 2) To probe robustness under varying shift intensities, we partition the dataset into multiple temporal subgraphs and train on one subgraph while testing on the others. Detailed statistics of each (sub)graph are provided in Appendix B.2.

Evaluation Metrics. We report two standard metrics: *Micro-F1* and *Macro-F1*. Micro-F1 is computed by first aggregating true/false positives and negatives over all classes, reflecting overall performance. Macro-F1 computes an F1 score for each class independently and then takes the unweighted mean, thereby giving equal importance to rare and frequent classes. All results are averaged over three random runs; we report the mean.

Baselines. We compare against representative methods from three categories. **(1) ERM.** Empirical Risk Minimization trains with the standard supervised loss under the i.i.d. assumption and serves as a basic reference. **(2) General OOD methods.** We include GroupDRO [34] and IRM [3], two widely used approaches for learning predictors that are invariant across environments in non-graph settings. **(3) Graph OOD methods (homogeneous).** We further consider EERM [42] and CaNet [41], which are designed for OOD generalization on homogeneous graphs. Since no prior work specifically targets heterogeneous graphs, we create heterogeneous adaptations by swapping their encoders for a heterogeneous backbone while keeping their original objectives unchanged. We denote the adapted variants as EERM* and CaNet*. All methods are implemented on heterogeneous graph backbones, including HeteroGCN and HeteroGraphSAGE. For simplicity, we abbreviate them as GCN and SAGE in the following experiments.

5.2 Performance Comparison (RQ1).

Evaluation under Degree-based Shifts. As shown in Table 2, on the Arxiv, Book, and Tmdb datasets with both SAGE and GCN backbones, InvHG achieves the highest average performance on OOD data, while maintaining performance on ID data that is comparable to or slightly better than the best baselines. This indicates that the improvements on OOD generalization are not obtained at the expense of ID accuracy. Compared with the strong graph generalization baseline CaNet*, InvHG consistently yields positive gains in average OOD Macro-F1 and Micro-F1. The most representative single improvement is observed on the GCN backbone with Arxiv, where the OOD Macro-F1 increases from 0.4680 to 0.524. In contrast to general OOD approaches such as ERM, GroupDRO, and IRM, the advantage of InvHG is even more pronounced, highlighting the benefit of its type- and relation-aware invariant learning in mitigating degree-related biases. Averaged across datasets, OOD improvements in Macro-F1 and Micro-F1 surpass their ID counterparts. Results under temporal shifts (Table 9), obtained by masking

Table 2: Performance comparison under degree-based OOD and ID settings.

Backbone	Model	Arxiv				Book				Tmdb				Avg			
		ID		OOD		ID		OOD		ID		OOD		ID		OOD	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.4498	0.8242	0.4118	0.6891	0.7068	0.7671	0.6651	0.7494	0.7205	0.7196	0.7232	0.7516	0.6257	0.7703	0.6000	0.7300
	GroupDRO	0.3901	0.7981	0.3605	0.6531	0.6829	0.7435	0.6469	0.7317	0.7088	0.7109	0.7164	0.7445	0.5939	0.7508	0.5746	0.7098
	IRM	0.4479	0.8277	0.4148	0.6902	0.6944	0.7606	0.6630	0.7508	0.7158	0.7159	0.7272	0.7556	0.6194	0.7681	0.6017	0.7322
SAGE	ERM	0.5210	0.8527	0.4741	0.7463	0.7620	0.8146	0.7374	0.8104	0.8245	0.8126	0.8125	0.8478	0.7025	0.8266	0.6747	0.8015
	GroupDRO	0.4553	0.8363	0.4203	0.7115	0.7621	0.8143	0.7375	0.8099	0.7972	0.7848	0.8060	0.8437	0.6715	0.8118	0.6546	0.7884
	IRM	0.4506	0.8406	0.4194	0.7282	0.7488	0.8057	0.7273	0.8054	0.8149	0.8022	0.8159	0.8513	0.6714	0.8162	0.6542	0.7950
	EERM*	0.4752	0.8183	0.5187	0.7185	OOM	OOM	OOM	OOM	0.8340	0.8312	0.8290	0.8623	—	—	—	—
	CANet*	0.4910	0.8533	0.4571	0.7405	0.7893	0.8336	0.7571	0.8225	0.8353	0.8225	0.8083	0.8476	0.7052	0.8365	0.6742	0.8035
	InvHG	0.5240	0.8580	0.4917	0.7488	0.8084	0.8506	0.7654	0.8261	0.8326	0.8261	0.8308	0.8641	0.7217	0.8449	0.6960	0.8130
GCN	ERM	0.4668	0.8427	0.4201	0.7284	0.7560	0.8101	0.7297	0.8095	0.8347	0.8222	0.8150	0.8540	0.6858	0.8250	0.6549	0.7973
	GroupDRO	0.4305	0.8293	0.4043	0.7048	0.7561	0.8094	0.7300	0.8084	0.8113	0.7978	0.8093	0.8483	0.6660	0.8122	0.6479	0.7872
	IRM	0.3996	0.8253	0.3364	0.6865	0.7408	0.8005	0.7169	0.8040	0.7984	0.7862	0.7320	0.8070	0.6463	0.8040	0.5951	0.7658
	EERM*	0.4499	0.8264	0.4225	0.6990	OOM	OOM	OOM	OOM	0.8064	0.8087	0.8110	0.8481	—	—	—	—
	CANet*	0.5072	0.8557	0.4680	0.7474	0.7738	0.8257	0.7472	0.8176	0.8178	0.8065	0.8141	0.8501	0.6996	0.8293	0.6764	0.8050
	InvHG	0.5381	0.8663	0.5240	0.7779	0.7961	0.8351	0.7627	0.8240	0.8405	0.8363	0.8277	0.8631	0.7249	0.8459	0.7048	0.8216

based on publication years, exhibit a similar trend and are reported in the Appendix D.1.

Evaluation with Multi-subgraph Partition under Varying Shifts. Unlike the degree-based mask in Table 2 or the two-subgraph partition in Table 10, multi-subgraph partition setup reflects a more realistic temporal rolling scenario, where the model is required to generalize from early structures to later and entirely unseen subgraphs. As reported in Table 3, InvHG achieves the best average OOD performance under both backbones. With SAGE, the average Macro-F1 and Micro-F1 are 0.4767 and 0.7904, surpassing the next best methods by 1–2 percentage points. Additionally, on Book (Table 4), InvHG attains the best performance across all temporal subgraphs, with Macro-F1 gains peaking at nearly 6 percentage points over the strongest baseline. Additional results on other datasets (Table 11) reveal consistent trends and are presented in the Appendix D.3. Taken together, these results indicate that InvHG maintains consistently strong OOD generalization in the multi-subgraph setting.

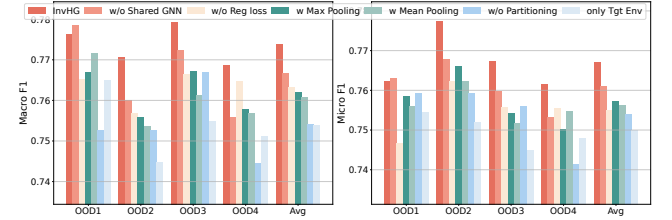
Overall, this experiment not only demonstrates the robustness of InvHG under transductive learning, but also highlights its ability to generalize to long-term unseen structures and semantic distributions. Across different partition strategies (time/degree-based masking, two/multi-subgraph), InvHG consistently achieves superior results, underscoring the method’s universality and stability. Furthermore, unlike other approaches that exhibit large performance fluctuations across different GNN encoders, InvHG maintains comparable performance under both SAGE and GCN backbones, showing strong robustness to encoder choice.

5.3 Ablation Study (RQ2)

We assess the contribution of each component in InvHG; results are summarized in Figure 4 under the multi-OOD subgraph setting. (i) *w/o Shared GNN*: removing the shared encoder for forward and reverse edges degrades both Macro-F1 and Micro-F1, indicating the benefit of parameter sharing for capturing directional symmetry. (ii) *w/o Reg loss*: dropping the KL regularization on environment vectors harms performance, showing that distributional consistency across environments is important for generalization. (iii) *w Max Pooling / w Mean Pooling*: replacing dynamic fusion with fixed max/mean pooling leads to noticeable drops, underscoring the advantage of adaptive fusion. (iv) *w/o Partitioning (CANet*)*: removing subgraph

partitioning reduces the model to CANet* with a heterogeneous encoder and consistently underperforms InvHG. (v) *only Tgt Env*: inferring environments only for target nodes yields the largest decline, as global distributional cues are not exploited.

Overall, InvHG achieves the best scores across subgraphs, with the largest gains attributed to global environment partitioning.

**Figure 4: Ablation study on TMDB under multi-OOD settings.**

5.4 Sensitivity Analysis (RQ3)

Impact of K . We first investigate the effect of the number of environments K (see Figure 5(a)). The overall trend shows that when K is within a moderate range (e.g., 3–7), the model achieves better performance on both Macro-F1 and Micro-F1, suggesting that the partitioning can effectively capture distributional variations across subgraphs. When K is too small (e.g., $K = 2$), the limited number of environments fails to represent diverse semantic shifts, leading to a drop in performance. On the other hand, when K is too large (e.g., $K = 9$ or 10), the overly fine-grained partitions introduce noise and sparsity, which slightly destabilizes the results. Nevertheless, the performance curves remain within a relatively stable range, demonstrating the robustness of our model with respect to K .

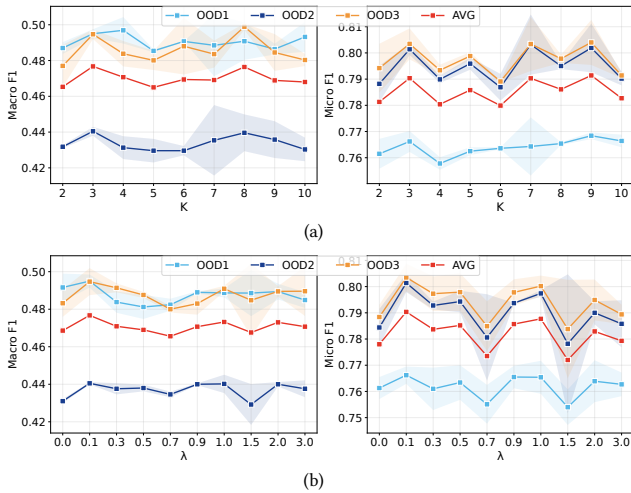
Impact of λ . We further examine the impact of the KL regularization weight λ (see Figure 5(b)). The results indicate that when λ is too small (e.g., 0–0.1), the constraint on environment vectors is insufficient, resulting in weakly separated environment representations and degraded performance. As λ increases to a moderate range (0.3–1.0), both Macro-F1 and Micro-F1 improve and become more stable, suggesting that a moderate KL divergence helps maintain discriminative and generalizable environment embeddings. However,

Table 3: Performance comparison under Arxiv Multi-OODs settings.

Backbone	Model	OOD1		OOD2		OOD3		Avg	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.3727	0.7270	0.3727	0.7270	0.4013	0.7244	0.3822	0.7261
	GroupDRO	0.3765	0.7330	0.3765	0.7330	0.3960	0.7271	0.3830	0.7310
	IRM	0.3620	0.7137	0.3620	0.7137	0.3859	0.7117	0.3700	0.7130
SAGE	ERM	0.5005	0.7569	0.4196	0.7667	0.4862	0.7697	0.4618	0.7644
	GroupDRO	0.4271	0.7774	0.4271	0.7774	0.4607	0.7797	0.4383	0.7782
	IRM	0.4045	0.7381	0.3945	0.7381	0.4454	0.7286	0.4148	0.7349
	EERM*	0.4136	0.7429	0.4136	0.7492	0.4593	0.7459	0.4288	0.7481
	CANet*	0.4224	0.7629	0.4224	0.7629	0.4533	0.7652	0.4327	0.7637
	InvHG	0.4995	0.7712	0.4405	0.8014	0.4947	0.8035	0.4782	0.7920
GCN	ERM	0.4524	0.7400	0.3950	0.7566	0.4186	0.7688	0.4220	0.7535
	GroupDRO	0.3986	0.7562	0.3986	0.7562	0.4262	0.7583	0.4078	0.7569
	IRM	0.3803	0.7327	0.3703	0.7327	0.3933	0.7279	0.3820	0.7311
	EERM*	0.4071	0.7445	0.3671	0.7445	0.3963	0.7422	0.3902	0.7437
	CANet*	0.4208	0.7555	0.4208	0.7555	0.4460	0.7561	0.4292	0.7557
	InvHG	0.4918	0.7506	0.4377	0.7819	0.4521	0.7856	0.4606	0.7727

Table 4: Performance comparison under Book Multi-OODs settings.

Backbone	Model	OOD1		OOD2		OOD3		OOD4		Avg	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.6274	0.7063	0.6274	0.7063	0.6092	0.6933	0.6063	0.6870	0.6176	0.6982
	GroupDRO	0.6270	0.7071	0.6270	0.7071	0.6075	0.6933	0.6055	0.6881	0.6163	0.6989
	IRM	0.5349	0.6523	0.5349	0.6523	0.5266	0.6463	0.5256	0.6401	0.5305	0.6477
SAGE	ERM	0.7153	0.7942	0.7153	0.7942	0.6993	0.7824	0.7019	0.7769	0.7080	0.7869
	GroupDRO	0.7134	0.7928	0.7134	0.7928	0.6897	0.7815	0.7001	0.7752	0.7062	0.7857
	IRM	0.6702	0.7682	0.6632	0.7662	0.6600	0.7503	0.6609	0.7242	0.6636	0.7522
	EERM*	0.6929	0.7789	0.6929	0.7789	0.6664	0.7645	0.6651	0.7309	0.6768	0.7633
	CANet*	0.7254	0.8000	0.7254	0.8000	0.7079	0.7870	0.7113	0.7821	0.7175	0.7923
	InvHG	0.7799	0.8148	0.7852	0.8088	0.7211	0.7914	0.7143	0.7860	0.7429	0.8003
GCN	ERM	0.6970	0.7874	0.6970	0.7874	0.6819	0.7791	0.6778	0.7687	0.6884	0.7806
	GroupDRO	0.6959	0.7866	0.6959	0.7866	0.6808	0.7782	0.6761	0.7674	0.6872	0.7797
	IRM	0.6602	0.7582	0.6582	0.7482	0.6600	0.7453	0.6469	0.7162	0.6563	0.7420
	EERM*	0.6691	0.7415	0.6691	0.7415	0.6794	0.7562	0.6503	0.7221	0.6695	0.7453
	CANet*	0.7329	0.8023	0.7329	0.8023	0.7137	0.7889	0.7113	0.7799	0.7227	0.7934
	InvHG	0.7823	0.8174	0.7478	0.8121	0.7262	0.7958	0.7256	0.7872	0.7455	0.8031

**Figure 5: Model performance analysis on the TMDb dataset under multi-OOD settings: (a) the number of environments K , and (b) the KL regularization weight λ .**

when λ is overly large (e.g., 1.5–3.0), the excessive regularization

over-separates environments and suppresses feature learning, leading to fluctuations in performance. Overall, InvHG consistently maintains near-optimal results across a wide range of hyperparameter settings, confirming the robustness of the method.

6 Conclusion

In this work, we present InvHG, a novel framework for OOD generalization on heterogeneous graphs, aimed at improving model's generalization under complex distribution shifts in real-world scenarios. The method combines environment partitioning with dynamic fusion to explicitly model unstable factors and aggregate stable relational patterns across subgraphs. Extensive evaluations across benchmark datasets and varied OOD settings, reveal that InvHG consistently outperforms sota baselines, and the ablation and sensitivity studies further validate its robustness and transferability. Overall, InvHG introduces a robust framework for heterogeneous graph learning, enabling resilient applications in web data, such as recommendation systems and social network analysis, under diverse distribution shifts.

Acknowledgments

This work was supported by the Science and Technology Innovation Program of Hunan Province (Grant No. 2024RC1041).

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A Theoretical Derivations

A.1 Derivation of the Interventional Objective (Single-E)

Setup and assumptions. Consider the causal graph with three variables $(E, G, \hat{Y})$ as in Figure 2, where E is a latent environment, G is the (typed) graph, and $\hat{Y}$ is the model prediction. The causal edges are $E \rightarrow G$, $G \rightarrow \hat{Y}$, and $E \rightarrow \hat{Y}$ (the last one arises through training under $p_{\text{tr}}(E)$). Let $\mathcal{A}$ be the corresponding DAG. For any node B , denote by $\mathcal{A}_{\leftarrow B}$ the graph obtained by cutting off all arrows into B , and by $\mathcal{A}_{B \rightarrow}$ the graph obtained by cutting off all arrows out of B . We will use two standard rules of do-calculus [33]:

- (i) *Action/observation exchange:* $P(d \mid do(b), do(e)) = P(d \mid do(b), e)$ if $(D \perp\!\!\!\perp E \mid B)_{\mathcal{A}_{\leftarrow BE}}$.
- (ii) *Insertion/deletion of actions:* $P(d \mid do(b), do(e)) = P(d \mid do(b))$ if $(D \perp\!\!\!\perp E \mid B)_{\mathcal{A}_{B \rightarrow E}}$.

Derivation. We aim to compute $p_\theta(\hat{Y} \mid do(G))$. By the law of total probability,

$$\begin{aligned} P(\hat{Y} \mid do(G)) &= \sum_e P(\hat{Y} \mid do(G), E=e) P(E=e \mid do(G)) \\ &\stackrel{(i)}{=} \sum_e P(\hat{Y} \mid G, E=e) P(E=e \mid do(G)) \\ &\stackrel{(ii)}{=} \sum_e P(\hat{Y} \mid G, E=e) P(E=e) \\ &= \mathbb{E}_{p_0(E)} \left[p_\theta(\hat{Y} \mid G, E) \right], \end{aligned} \quad (16)$$

where the first step uses total probability, the second step uses rule (i) because $\hat{Y} \perp\!\!\!\perp E \mid G$ in $\mathcal{A}_{\leftarrow G}$, and the third step uses rule (ii) because $E \perp\!\!\!\perp do(G)$ in $\mathcal{A}_{G \rightarrow}$. Here $p_0(E)$ is a prior over environments. This yields the interventional objective used in the main text.

A.2 Variational Lower Bound for Equation (16)

To obtain a tractable learning objective, we introduce a variational estimator $q_\phi(E \mid G)$ and apply Jensen's inequality:

$$\begin{aligned} \log p_\theta(\hat{Y} \mid do(G)) &= \log \sum_e p_\theta(\hat{Y} \mid G, E=e) p_0(E=e) \\ &= \log \sum_e p_\theta(\hat{Y} \mid G, E=e) p_0(E=e) \frac{q_\phi(E=e \mid G)}{q_\phi(E=e \mid G)} \\ &\geq \mathbb{E}_{q_\phi(E \mid G)} \left[\log p_\theta(\hat{Y} \mid G, E) \right] - \text{KL}(q_\phi(E \mid G) \parallel p_0(E)). \end{aligned} \quad (17)$$

This is the variational lower bound adopted in the main paper.

A.3 Relation-wise Extension for HG

Type-aware environments. In heterogeneous graphs, different relation types $r \in \mathcal{R}$ may be subject to distinct environmental factors. We therefore consider a collection of latent environments $\mathbf{E} = \{E^r\}_{r \in \mathcal{R}}$ and the family of typed subgraphs $\{G^{(r)}\}_{r \in \mathcal{R}}$. Assuming E^r are independent under the prior,¹ $p_0(\mathbf{E}) = \prod_{r \in \mathcal{R}} p_0(E^r)$, we obtain

$$p_\theta(\hat{Y} \mid do(G)) = \mathbb{E}_{p_0(\mathbf{E})} \left[p_\theta(\hat{Y} \mid \{G^{(r)}, E^r\}_{r \in \mathcal{R}}) \right]. \quad (18)$$

¹This is a standard simplifying assumption; relaxing it would replace the product prior by a structured prior.

Variational bound (relation-wise). Introduce a factorized variational family $q_\phi(\mathbf{E} \mid G) = \prod_{r \in \mathcal{R}} q_\phi(E^r \mid G^{(r)})$. Then

$$\begin{aligned} \log p_\theta(\hat{Y} \mid do(G)) &= \log \int p_\theta(\hat{Y} \mid \{G^{(r)}, E^r\}) \prod_r p_0(E^r) d\mathbf{E} \\ &= \log \int p_\theta(\hat{Y} \mid \{G^{(r)}, E^r\}) \prod_r \frac{p_0(E^r)}{q_\phi(E^r \mid G^{(r)})} \prod_r q_\phi(E^r \mid G^{(r)}) d\mathbf{E} \\ &\geq \mathbb{E}_{\prod_r q_\phi(E^r \mid G^{(r)})} \left[\log p_\theta(\hat{Y} \mid \{G^{(r)}, E^r\}) \right] - \sum_{r \in \mathcal{R}} \text{KL}(q_\phi(E^r \mid G^{(r)}) \parallel p_0(E^r)). \end{aligned} \quad (19)$$

Thus the multi-relation ELBO naturally decomposes into a prediction term and a *sum of relation-wise KL regularizers*, exactly matching the training loss used in Sec. 4.3.

From latent environments to pseudo-environment embeddings. In practice we instantiate $q_\phi(E^r \mid G^{(r)})$ by producing logits $\pi_v^{[l],r}$ for the destination node v and sampling a (soft) categorical pseudo-environment embedding $\mathbf{e}_v^{[l],r}$ using the Gumbel-Softmax trick. With a uniform categorical prior $p_0(E^r) = \text{Cat}(\frac{1}{K}, \dots, \frac{1}{K})$, the KL admits the standard surrogate

$$\text{KL}(q_\phi(E^r \mid G^{(r)}) \parallel p_0(E^r)) \approx \sum_{k=1}^K e_{vk}^{[l],r} (\log \pi_{vk}^{[l],r} + \log K),$$

yielding the objective in the main paper.

B Datasets Details

B.1 Overall of Datasets

The statistics of the original dataset are summarized in Table 5. For brevity, we denote task type as T-type, multi-class as MC, multi-label as ML, number of classes as Cls, and target node as T-node; edge types are abbreviated, e.g., Movie-Actor (M-A), Movie-Director (M-D), Paper-Paper (P-P), Paper-Author (P-A), Paper-FoS (P-F), Book-Book (B-B), Book-Author (B-A), Book-Publisher (B-P).

Table 5: Statistics of Datasets.

Dataset	Node		Edge		Cls	T-type	T-node
TMDB	Movie	7,505	M-A	86,517	4	MC	Movie
	Actor	13,016	M-D	18,341			
	Director	3,891	-	-			
		24,412		104,858			
ArXiv	Paper	81,634	P-P	1,019,624	40	MC	Paper
	Author	127,590	P-A	300,233			
	FoS	21,887	P-F	755,835			
		231,111		2,075,692			
Book	Book	594,484	B-B	7,614,902	8	ML	Book
	Author	147,863	B-A	825,905			
	Publisher	43,910	B-P	594,484			
		786,257		9,035,291			

ArXiv (CS) is an academic network constructed from arXiv Computer Science and the Microsoft Academic Graph [18], consisting of paper, author, and field-of-study nodes, where the task is 40-class paper node classification. Book is a heterogeneous literature network derived from the Goodreads Book Graph, involving books, authors, and publishers, with a multi-label book classification task over eight genres. TMDB is a heterogeneous movie network collected from The Movie Database API, containing movies, actors, and directors, where the task is 4-class movie genre classification. For all datasets, only the target nodes (papers, books, or movies) are associated with 384-d MiniLM text embeddings, while the remaining node types are textless.

B.2 Multi-subgraph temporal splits.

To expose models to *diverse* distribution shifts, each dataset is partitioned into multiple temporal subgraphs (Train, OOD1, OOD2, ...). We enforce two constraints during partitioning: (i) the target-node counts across subgraphs are kept roughly comparable; (ii) the target nodes' years are non-overlapping across subgraphs. This construction yields multiple OOD test environments with different densities and relation compositions, as reflected by the varying edge totals. Detailed statistics are listed in Tables 6, 7 and Tables 8.

ArXiv. Publication years are partitioned into disjoint intervals: Train (1986–2014), OOD1 (2015–2017), OOD2 (2018), and OOD3 (2019–2020), with the number of target papers kept roughly comparable across splits.

Table 6: Statistics of subgraph splits for ArXiv dataset.

Subgraph	Node			Edges		
	Paper	Author	FoS	P-P	P-A	P-F
Train	18176	28368	12466	25013	54449	169957
OOD1	28908	53324	14501	78403	106586	274906
OOD2	18170	45197	11048	14889	72819	164063
OOD3	16380	45059	10203	8597	66953	146909

Book. The corpus is segmented into non-overlapping year ranges: Train (<2003), OOD1 (2004–2009), OOD2 (2010–2012), OOD3 (2013–2014), and OOD4 (2015–2023)—choosing boundaries to balance target-book counts.

Table 7: Statistics of subgraph splits for Book dataset.

Subgraph	Node			Edges		
	Book	Author	Publisher	B-B	B-A	B-P
Train	129293	42196	10980	417596	185134	129293
OOD1	120833	47128	8212	292116	177073	120833
OOD2	135940	51797	14261	422062	187519	135940
OOD3	122577	52373	16257	419878	162434	122577
OOD4	82762	37891	10263	242196	110338	82762

TMDB. Release years are stratified into mutually exclusive intervals: Train (<1977), OOD1 (1978–1999), OOD2 (2000–2010), OOD3 (2011–2017), and OOD4 (2018–2025), aiming for approximately equal target-movie counts per subgraph.

Table 8: Statistics of subgraph splits for TMDB dataset.

Subgraph	Node			Edges	
	Movie	Actor	Director	M-A	M-D
Train	1511	2910	540	14023	2617
OOD1	1532	6012	1418	19209	3900
OOD2	1612	7621	1968	22845	4537
OOD3	1502	6935	1967	18201	4005
OOD4	1313	5483	1661	12174	3234

C Experimental Details

All experiments were conducted on a single NVIDIA RTX 3090 workstation with 32GB system RAM, using Python 3.11.13 with PyTorch 2.3.1 and PyG 2.6.1, and trained with the Adam [22] optimizer with weight decay. Beside, We perform hyper-parameter optimization for each dataset using a grid-style search on the validation set. The search spaces are defined as follows:

- Number of GNN layers L : {1, 2, 3, 4}.
- Learning rate: {0.001, 0.005, 0.01}.
- Regularization weight λ : sampled uniformly from [0.1, 1.0] with step size 0.1.
- Gumbel-Softmax temperature τ_g : {0.0, 0.1, 0.2, 0.5, 1.0, 2.0, 3.0, 4.0}.
- Number of pseudo environments K : {2, 3, 4, 5, 6, 7}.

D Additional Results under Variant Distribution Shifts

D.1 Evaluation under Temporal Shifts

We further evaluate within-graph temporal shifts by masking nodes according to publication years to form ID and OOD splits. As reported in Table 9, across Arxiv, Book, and Tmdb with both SAGE and GCN backbones, InvHG maintains the top average performance on OOD Macro-F1 and Micro-F1 while keeping ID results on par with the best baselines. Using GCN as an example, the average OOD scores improve from CaNet*'s 0.6895 (Macro-F1) and 0.8235 (Micro-F1) to 0.7057 and 0.8369, respectively, i.e., +1.62 and +1.34 percentage points. With SAGE, the average OOD scores increase from 0.6843 and 0.8168 to 0.7031 and 0.8300, i.e., +1.88 and +1.32 percentage points. Overall, these time-mask results mirror the degree-mask trend in the main text, reinforcing the robustness of InvHG to temporal distribution shifts.

D.2 Evaluation under Two-Subgraph Partition.

Table 10 shows that InvHG consistently achieves superior performance on Arxiv, Book, and Tmdb, with OOD Macro-F1 and Micro-F1 leading across different backbones. On average, the OOD Macro-F1 reaches 0.6928 with SAGE and 0.6867 with GCN, representing improvements of 1-3 percentage points compared to CaNet*. Compared to the ERM, the maximum improvement reaches 4.4%. These findings demonstrate that InvHG can effectively adapt to distributional discrepancies across subgraphs and exhibits robust transferability in cross-subgraph generalization scenarios.

D.3 Evaluation with Multi-subgraph Partition on Tmdb

The results in Table 11 indicate that, regardless of whether SAGE or GCN is used as the backbone, InvHG consistently achieves superior average OOD performance and maintains advantages on most OOD subgraphs, demonstrating the model's consistent cross-dataset effectiveness and robustness to encoder selection.

Table 9: Performance comparison under time-based OOD and ID settings.

Backbone	Model	Arxiv				Book				Tmdb				Avg			
		ID		OOD		ID		OOD		ID		OOD		ID		OOD	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.4956	0.7409	0.4336	0.7907	0.7068	0.7671	0.6651	0.7494	0.7456	0.7602	0.7068	0.7056	0.6493	0.7561	0.6018	0.7486
	GroupDRO	0.4287	0.7054	0.3775	0.7605	0.6829	0.7435	0.6469	0.7317	0.7243	0.7375	0.6808	0.6764	0.6120	0.7288	0.5684	0.7229
	IRM	0.4978	0.7392	0.4418	0.7901	0.6944	0.7606	0.6630	0.7508	0.7587	0.7714	0.7106	0.7088	0.6503	0.7571	0.6051	0.7499
SAGE	ERM	0.5430	0.7929	0.4943	0.8348	0.7620	0.8147	0.7375	0.8104	0.8322	0.8469	0.8195	0.8096	0.7124	0.8182	0.6838	0.8183
	GroupDRO	0.4984	0.7664	0.4437	0.8177	0.7621	0.8142	0.7373	0.8099	0.8104	0.8267	0.7938	0.7820	0.6903	0.8024	0.6583	0.8032
	IRM	0.4856	0.7742	0.4412	0.8235	0.7488	0.8058	0.7273	0.8054	0.8207	0.8370	0.8131	0.8025	0.6850	0.8057	0.6605	0.8105
	EERM*	0.5100	0.7374	0.4875	0.8072	OOM	OOM	OOM	OOM	0.8243	0.8422	0.8322	0.8286	-	-	-	-
	CANet*	0.5607	0.7945	0.4940	0.8335	0.7765	0.8265	0.7484	0.8176	0.8198	0.8363	0.8106	0.7994	0.7190	0.8191	0.6843	0.8168
	InvHG	0.5913	0.8139	0.5211	0.8502	0.8025	0.8400	0.7565	0.8169	0.8485	0.8636	0.8317	0.8231	0.7474	0.8392	0.7031	0.8300
GCN	ERM	0.5104	0.7751	0.4623	0.8262	0.7561	0.8101	0.7298	0.8095	0.8282	0.8438	0.8185	0.8078	0.6982	0.8097	0.6702	0.8145
	GroupDRO	0.4723	0.7526	0.4173	0.8137	0.7562	0.8094	0.7299	0.8084	0.7856	0.8001	0.8002	0.7884	0.6714	0.7874	0.6491	0.8035
	IRM	0.4012	0.7411	0.3872	0.8066	0.7408	0.8005	0.7169	0.8040	0.8131	0.8289	0.7875	0.7787	0.6517	0.7902	0.6305	0.7964
	EERM*	0.4863	0.7633	0.4538	0.8136	OOM	OOM	OOM	OOM	0.7961	0.8142	0.8143	0.8101	-	-	-	-
	CANet*	0.5497	0.7964	0.4916	0.8389	0.7888	0.8334	0.7571	0.8226	0.8280	0.8451	0.8198	0.8091	0.7222	0.8250	0.6895	0.8235
	InvHG	0.6036	0.8088	0.5145	0.8551	0.8005	0.8433	0.7652	0.8278	0.8526	0.8658	0.8375	0.8275	0.7522	0.8393	0.7057	0.8368

Table 10: Performance under Two-Subgraph Partition.

Backbone	Model	Arxiv		Book		Tmdb		Avg	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.4341	0.7605	0.6615	0.7356	0.6958	0.7047	0.5971	0.7336
	GroupDRO	0.4297	0.7537	0.6578	0.7326	0.6896	0.6985	0.5924	0.7283
	IRM	0.4104	0.7371	0.5558	0.6721	0.6986	0.7079	0.5549	0.7057
SAGE	ERM	0.4783	0.8010	0.7345	0.8052	0.7939	0.7939	0.6689	0.8000
	GroupDRO	0.4845	0.8083	0.7339	0.8044	0.8104	0.8088	0.6763	0.8072
	IRM	0.4661	0.7904	0.7202	0.7902	0.7944	0.7958	0.6602	0.7921
	EERM*	0.4625	0.7989	0.7221	0.8002	0.7971	0.7850	0.6606	0.7947
	CANet*	0.4898	0.8047	0.7357	0.8056	0.7936	0.7924	0.6730	0.8009
	InvHG	0.4968	0.8090	0.7667	0.8225	0.8149	0.8130	0.6928	0.8148
GCN	ERM	0.4344	0.7860	0.7226	0.8006	0.8061	0.8073	0.6544	0.7980
	GroupDRO	0.4392	0.7903	0.7212	0.7998	0.8133	0.8143	0.6579	0.8015
	IRM	0.3925	0.7599	0.7001	0.7901	0.7829	0.7832	0.6252	0.7777
	EERM*	0.4177	0.7652	0.7147	0.7963	0.7797	0.7828	0.6374	0.7814
	CANet*	0.4846	0.8040	0.7425	0.8087	0.7814	0.7805	0.6695	0.7977
	InvHG	0.4899	0.8084	0.7666	0.8236	0.8176	0.8155	0.6914	0.8158

Table 11: Performance comparison under TMDB Multi-OODs settings.

Backbone	Model	OOD1		OOD2		OOD3		OOD4		Avg	
		Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1	Macro F1	Micro F1
MLP	ERM	0.6391	0.6736	0.6391	0.6736	0.6244	0.6358	0.5895	0.6087	0.6230	0.6479
	GroupDRO	0.6329	0.6682	0.6329	0.6682	0.6196	0.6305	0.5827	0.6006	0.6170	0.6419
	IRM	0.6419	0.6787	0.6419	0.6787	0.6431	0.6520	0.6033	0.6184	0.6325	0.6569
SAGE	ERM	0.7539	0.7591	0.7539	0.7591	0.7619	0.7501	0.7287	0.7254	0.7496	0.7485
	GroupDRO	0.7494	0.7581	0.7494	0.7581	0.7573	0.7462	0.7227	0.7197	0.7447	0.7455
	IRM	0.7550	0.7622	0.7550	0.7622	0.7636	0.7519	0.7314	0.7284	0.7512	0.7512
	EERM*	0.7253	0.7492	0.7253	0.7492	0.7338	0.7308	0.7189	0.7218	0.7258	0.7378
	CANet*	0.7527	0.7593	0.7527	0.7593	0.7668	0.7559	0.7444	0.7413	0.7542	0.7539
	InvHG	0.7763	0.7662	0.7705	0.7773	0.7793	0.7674	0.7486	0.7616	0.7737	0.7671
GCN	ERM	0.7725	0.7744	0.7725	0.7744	0.7615	0.7488	0.7526	0.7493	0.7648	0.7617
	GroupDRO	0.7744	0.7787	0.7744	0.7787	0.7571	0.7446	0.7465	0.7452	0.7631	0.7618
	IRM	0.7553	0.7593	0.7553	0.7593	0.7413	0.7277	0.7270	0.7228	0.7447	0.7423
	EERM*	0.7569	0.7587	0.7569	0.7587	0.7587	0.7461	0.7488	0.7446	0.7553	0.7520
	CANet*	0.7573	0.7686	0.7573	0.7686	0.7603	0.7523	0.7364	0.7370	0.7528	0.7566
	InvHG	0.7817	0.7672	0.7774	0.7812	0.7716	0.7617	0.7507	0.7474	0.7704	0.7644